WORKSHEET 2.5

Power, Sum and Constant-Multiple Rules

Unit 2: The Derivative

Key ideas

- $\dfrac{d}{dx}\left[x^n\right] = nx^{n-1}$ for every real number n.
- Constants multiply through, and sums are differentiated term by term. The derivative of a constant is 0.
- **Rewrite first**: $\sqrt{x} = x^{1/2}$, $\dfrac{1}{x^3} = x^{-3}$. Expand products and divide out simple quotients.
- A horizontal tangent occurs where $f'(x) = 0$; a tangent parallel to a line of slope m occurs where $f'(x) = m$.

Worked examples

Example 1 · Term by term

Find $\dfrac{d}{dx}\left[2x^6 - 5x^3 + 4x - 1\right]$.

Solution. Apply the power rule to each term.

$12x^5 - 15x^2 + 4$.

Example 2 · Rewriting

Find $\dfrac{d}{dx}\left[\dfrac{6}{x^3} - 2\sqrt{x}\right]$.

Solution. Write as $6x^{-3} - 2x^{1/2}$, giving $-18x^{-4} - x^{-1/2}$.

$-\dfrac{18}{x^4} - \dfrac{1}{\sqrt{x}}$.

Example 3 · A tangent line

Write an equation of the line tangent to $y = x^4 - 2x$ at $x = 1$.

Solution. $y(1) = -1$ and $y' = 4x^3 - 2$, so the slope is 2.

$y + 1 = 2(x - 1)$.

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Practice

CORE

1. Find $\frac{d}{dx}\left[x^7\right]$.

2. Find $\frac{d}{dx}\left[3x^4 - x + 5\right]$.

3. Find $\frac{d}{dx}\left[10\sqrt{x}\right]$.

4. Find $\frac{d}{dx}\left[\frac{2}{x}\right]$.

EXAM LEVEL

5. Find $f'(x)$ for $f(x) = \left(x^2 + 1\right)(x - 3)$ by expanding first.

6. Find $f'(x)$ for $f(x) = \frac{x^3 - 4x}{x}$, $x \neq 0$.

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7. Find the coordinates of the points on $y = 2x^3 - 24x$ at which the tangent line is horizontal.

8. The function $f(x) = ax^3 + bx^2$ satisfies $f(1) = 5$ and $f'(1) = 12$. Find a and b.

Question 9 · Free-response practice

Let $p(x) = x^3 - 3x^2 - 24x + 5$.

(a) Find $p'(x)$.

(b) Find all values of x at which the tangent line to the graph of p is horizontal.

(c) Write an equation of the tangent line at $x = 0$, and find the other value of x at which the tangent line is parallel to it.

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Worked solutions

1. $7x^6$.
2. $12x^3 - 1$.
3. $5x^{-1/2} = \dfrac{5}{\sqrt{x}}$.
4. $-2x^{-2} = -\dfrac{2}{x^2}$.
5. $f(x) = x^3 - 3x^2 + x - 3$, so $f'(x) = 3x^2 - 6x + 1$.
6. $f(x) = x^2 - 4$, so $f'(x) = 2x$.
7. $y' = 6x^2 - 24 = 0$ at $x = \pm 2$: the points $(2, -32)$ and $(-2, 32)$.
8. $a + b = 5$ and $3a + 2b = 12$. Then $a = 2$ and $b = 3$.
9. (a) $p'(x) = 3x^2 - 6x - 24$.
   (b) $3(x - 4)(x + 2) = 0$, so $x = 4$ and $x = -2$.
   (c) $p(0) = 5$ and $p'(0) = -24$: $y = -24x + 5$. $3x^2 - 6x - 24 = -24$ gives $3x(x - 2) = 0$, so the other value is $x = 2$.

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